

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO5: *Create graphical models to analyze and infer patterns in complex probabilistic systems for structured decision-making.(BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:50 Annexure A

Mock Interview

Code of Conduct:

I ___PradeepKumar(192425016)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

*action. **Signature of the student:***

Annexure A

Mock Interview

Q1. "Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?" (5 Marks – 8 Mins)

1. I will Draw the Map, Draw Circles for the things we can measure like workload, Age, Temp, etc.
(ii) I will assign probabilities. Ex. If a machine is old and running hot, there's a 90% chance it will fail.

Q4. "During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?" (5 Marks – 8 Mins)

4. I will Build word Connections first: Line up all words in sentence and then I will create simple rules to identify the Grammar and I will Score the sentence.

Q6. "In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?" (5 Marks – 8 Mins)

6. In modelling physical diseases, I will choose directed graph because they naturally capture biological cause and effect. If modelling syndromes or mental health I will choose undirected graph.

Q9. "In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?" (5 Marks – 8 Mins)

9. HMMs are forced to assume that each word is completely independent of the rest of the sentence once you know its label. CRFs don't care about generating words; they only care about assigning labels.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively

Confidence	15	Displays high confidence, positive attitude, and professional	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
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behavior throughout

Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately
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